

Polynomial Zeros and Turning-Point Constraints

Answer Key

Precalculus Polynomial and Rational Functions

Quick Answers

- | | |
|-----------------------------|--|
| 1. 2, 5 | 7. 0, 3 |
| 2. $x(x-2)(x+2)$, 0, 2, -2 | 8. 5 |
| 3. 3, -1 | 9. $(x-2)(x-1)(x+1)(x+2)$, 1, -1, 2, -2 |
| 4. 4, 5 | 10. -2, 1, 3 |
| 5. 3 | 11. 2 |
| 6. 1, -2, 3 | 12. — |
-

Curriculum

Level: Precalculus Polynomial and Rational Functions

HSA-APR.B.3 – *problems 1, 2, 3, 6, 7, 9, 10*

HSA-APR.D.6–D.7 – *problems 5, 11*

HSF-IF.A–HSF-IF.C – *problems 4, 8, 12*

Difficulty 1–5 of 5 across 16 tagged checks.

1. $f(x) = x^2 - 7x + 10$

Step 1. Zeros are the solutions of $f(x) = 0$, and the polynomial is monic, so look for a factor pair of 10 that adds to -7 : those are -2 and -5 .

Step 2. $f(x) = (x-2)(x-5)$, so the zero product property gives the two zeros. $x = 2, 5$

2. $g(x) = x^3 - 4x$

(a) Step 1. Always take the greatest common factor first: $g(x) = x(x^2 - 4)$.

Step 2. The second factor is a difference of squares: $x^2 - 4 = (x-2)(x+2)$.

(b) Set each factor to zero. Three linear factors, three distinct zeros — the maximum a cubic can have.

$x(x-2)(x+2); \quad x = 0, \pm 2$

3. $p(x) = (x-3)^2(x+1)$

Step 1. The function is already factored, so read the zeros and their exponents directly: $x = 3$ from $(x-3)^2$ and $x = -1$ from $(x+1)$.

Step 2. The exponent on a factor *is* the multiplicity of that zero: multiplicity 2 at $x = 3$, multiplicity 1 at $x = -1$.

Step 3. Even multiplicity means the factor never changes sign there, so the graph touches and turns back; odd multiplicity means it changes sign, so the graph crosses.

$x = 3 \text{ (} m = 2, \text{ touch)}, \quad x = -1 \text{ (} m = 1, \text{ cross)}$

4. Degree-5 polynomial h .

(a) **Step 1.** A polynomial of degree n has at most $n - 1$ turning points, because between consecutive turning points the graph is monotonic and the derivative — a degree-4 polynomial here — has at most 4 roots.

(b) **Step 2.** It has at most n real zeros, since a degree- n polynomial has exactly n roots counted with multiplicity over the complex numbers, and real ones are a subset.

Both are ceilings. $h(x) = x^5$ has degree 5 with one zero and no turning points at all.

4 turning points; 5 zeros

5. $\lim_{x \rightarrow \infty} \frac{3x^2 - 5}{x^2 + 4}$

Step 1. Numerator and denominator have the *same* degree, so divide every term by the highest power, x^2 :

$$\frac{3 - \frac{5}{x^2}}{1 + \frac{4}{x^2}}.$$

Step 2. As $x \rightarrow \infty$ both fraction terms go to 0, leaving the ratio of the leading coefficients, $3/1$.

Step 3. A finite limit at infinity is a horizontal asymptote.

$y = 3$

6. $f(x) = x^3 - 2x^2 - 5x + 6$

Step 1. By the Rational Root Theorem any rational zero divides the constant term 6, so test $\pm 1, \pm 2, \pm 3, \pm 6$. At $x = 1$: $1 - 2 - 5 + 6 = 0$, so $(x - 1)$ is a factor.

Step 2. Divide: $f(x) = (x - 1)(x^2 - x - 6)$.

Step 3. Factor the quadratic: $x^2 - x - 6 = (x - 3)(x + 2)$.

Step 4. Three distinct real zeros, so the graph crosses three times and must have exactly 2 turning points.

$x = 1, -2, 3$

7. $q(x) = x^4 - 6x^3 + 9x^2$

Step 1. Common factor first: $q(x) = x^2(x^2 - 6x + 9)$.

Step 2. The bracket is a perfect square: $x^2 - 6x + 9 = (x - 3)^2$, so $q(x) = x^2(x - 3)^2$.

Step 3. Two distinct zeros, each of multiplicity 2. Both multiplicities are even, so the graph *touches* at both and crosses the x -axis **zero** times — the whole graph sits on or above the axis.

$x = 0$ ($m = 2$), $x = 3$ ($m = 2$)

8. A graph with exactly 4 turning points.

Step 1. Run the bound backwards. Degree n allows at most $n - 1$ turning points, so 4 turning points require $n - 1 \geq 4$, that is $n \geq 5$.

Step 2. Degree 5 is achievable — a quintic with four turning points exists — so the smallest possible degree is exactly 5.

Expected explanation: “A degree- n polynomial has at most $n - 1$ turning points, so 4 turning points need $n - 1 \geq 4$.”

5

9. $r(x) = x^4 - 5x^2 + 4$

(a) Step 1. This is quadratic in x^2 . With $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$, so $r(x) = (x^2 - 1)(x^2 - 4)$.

Step 2. Each bracket is a difference of squares, so factor again — “completely” means down to linear factors here.

(b) Step 3. Four distinct real zeros. A quartic can have at most 3 turning points, and four crossings force a turn between each consecutive pair, so it has exactly 3.

$(x - 1)(x + 1)(x - 2)(x + 2); x = \pm 1, \pm 2; 3 \text{ turning points}$

10. $s(x) = (x + 2)(x - 1)^2(x - 3)$

(a) Step 1. Read the zeros and exponents from the factored form: $x = -2$ (multiplicity 1), $x = 1$ (multiplicity 2), $x = 3$ (multiplicity 1).

(b) Step 2. Sketch checklist — this is what to look for in the student’s graph:

- Crosses the x -axis at $x = -2$ and at $x = 3$ (odd multiplicity).
- Touches and turns back at $x = 1$ without crossing (even multiplicity).
- Degree 4 with positive leading coefficient, so both ends rise.
- Three turning points — the maximum for a quartic — one of which is the touch point at $x = 1$.
- Below the axis on $(-2, 1)$ and on $(1, 3)$; above it outside $[-2, 3]$.

The sketch itself is flagged for manual review: judge it on the five bullets above, not on exact y -values.

$x = -2, 1 (m = 2), 3$

11. $\lim_{x \rightarrow \infty} \frac{4x^2 + 3}{2x^2 - x}$

Step 1. Equal degrees again, so divide numerator and denominator by x^2 :

$$\frac{4 + \frac{3}{x^2}}{2 - \frac{1}{x}}.$$

Step 2. Both fraction terms vanish as $x \rightarrow \infty$, leaving $4/2$.

Step 3. So the graph levels off at the horizontal asymptote below. (The vertical asymptotes at $x = 0$ and $x = \frac{1}{2}$ do not affect end behaviour.)

$y = 2$

12. Why 5 distinct real zeros and 2 turning points is impossible.

Model answer. Suppose the graph crosses the x -axis at five different places. Between two consecutive x -intercepts the function is zero at both ends and nonzero in between, so it must rise and then come back down (or fall and come back up) — it has to turn at least once in each of those gaps. Five distinct zeros create four gaps, so the graph has at least 4 turning points, not 2. A degree-5 polynomial is allowed at most 4 turning points, so four is exactly what such a polynomial must have — and 2 is impossible.

Grading note: this is a justification, so it is flagged for manual review rather than machine-checked. Full credit needs the “must turn between consecutive zeros” idea; naming Rolle’s theorem is not required.